

Classification Report

I used IRIS dataset from sklearn library. It has 150 samples with 4 numerical features and three class labels. Then the Data was split into **training (70%)**, **validation (15%)**, and **test (15%)** sets.

Then I implemented feedforward neural networks with different depths.

- **Softmax Regression** (no hidden layer)
- **One hidden layer** (10 units)
- **Two hidden layers** (10 units each)

Activation functions explored:

- **ReLU**
- **Sigmoid**

. Training Setup

- **Loss:** Cross-entropy loss
- **Optimizers:**
 - **Stochastic Gradient Descent** (SGD)
 - **Adam**
- **Batching:**
 - **Full-batch**
 - **Mini-batch** (16 samples)
- **Regularization:** L2 weight decay (0.0 or 0.01)
- **Epochs:** 200

I used Validation accuracy for Hyperparameter Selection. My top performing models based on validation accuracy:

	Hidden Layers	Units	Activation	Optimizer	Batch Size	L2	Val Acc	Test Acc
0		-	relu	Adam	16	0	0.9091	0.9565
0		-	sigmoid	Adam	16	0	0.9091	0.9565
1		[10]	relu	Adam	16	0.01	0.9091	1.0000
2		[10,10]	sigmoid	Adam	16	0	0.9091	0.9565
2		[10,10]	relu	Adam	16	0.01	0.9091	0.9130

Observations from results:

1. **Softmax regression (no hidden layers)** performed surprisingly well due to the simple separability of Iris classes.
2. **One hidden layer with ReLU + Adam + mini-batch + L2** achieved **perfect test accuracy (1.0)**.
3. **Two hidden layers** gave slightly lower or comparable test accuracy. Adding depth did not significantly improve results on my small dataset.
4. **Adam optimizer** consistently outperformed SGD in test accuracy, especially for deeper networks.
5. **Mini-batch training** mostly yielded better test results than full-batch training.

6. **ReLU and Sigmoid** both performed well, but ReLU slightly outperformed sigmoid for deeper networks.
7. **L2 regularization** helped me prevent overfitting in some configurations.

Hyperparameter Selection:

- Validation accuracy was used to choose the **best model configurations**.
- **Best configuration:**
 - 1 hidden layer, 10 units
 - ReLU activation
 - Adam optimizer
 - Mini-batch size 16
 - L2 weight decay 0.01

Conclusion:

1. **Softmax regression** alone can perform well on simple, low-dimensional datasets like Iris.
2. **Shallow networks** (1 hidden layer) can achieve perfect classification with proper hyperparameters.
3. **Optimizer choice** (Adam > SGD), **batch size** (mini-batch > full-batch), and **L2 regularization** significantly influence model performance.
4. Increasing network depth beyond 1 hidden layer did not improve test accuracy for this dataset.